Reading EPC charts

Event-driven Process Chains · one-page cheat sheet for the Pepa use-case charts

event → function → event

Event

A passive state, before or after work. It never decides and never acts.

Word it: noun + past tense, “Wake word heard”

Function

An active task that turns one state into the next.

Word it: verb + object, “Parse the command”

Connector

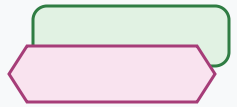
AND ∧ · XOR × · OR ∨. Splits one flow, or joins several.

Word it: no words, only the symbol

Control flow

The order things happen in. It says nothing about how long.

Word it: unlabeled, except yes / no



Process interface

A hand-off: another EPC chart picks up from here.

Word it: the other process’s name

CONNECTORS: SPLIT (1 IN) OR JOIN (1 OUT)

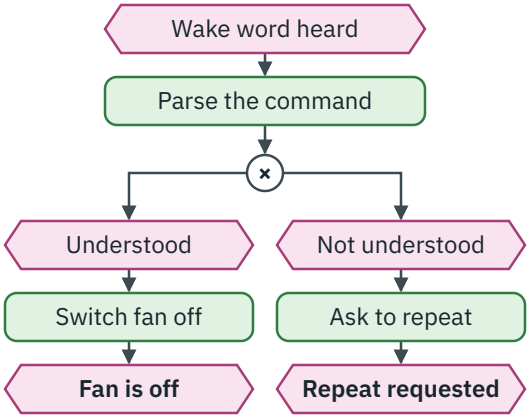
Symbol	As a split	As a join
∧ AND	All branches start, in parallel	Wait until every branch finishes
× XOR	Exactly one branch is taken	Go on when the one branch arrives
∨ OR	One or more branches	Wait for all that started

Events cannot decide. An XOR or OR split always follows a function; only AND may follow an event.

RULES OF A WELL-FORMED CHART

1. Every path starts on an event and ends on an event.
2. Events and functions alternate; connectors in between do not count.
3. No arrow from event to event, or from function to function.
4. Each split closes with a join of the same type.
5. A function that can fail shows its failure event too, e.g. “Nothing found” (Pepa practice).

WORKED EXAMPLE · UC4 FAN



READ ANY CHART IN 7 STEPS

1. Find the start events: no arrow in.
2. Find the end events: one per outcome.
3. Walk the happy path aloud.
4. Check the alternation along each path.
5. At each split, the function before it decides.
6. Match every split to its join.
7. Last, read who does it and what data it uses.

TRAPS

- Reading an event as an action
- An XOR right after an event
- OR join: ask which branches really ran
- Arrows show order, never time

Sources: Mendling, Neumann & Nüttgens (2005), “Yet Another Event-driven Process Chain,” EMISA 1(1) · Bruni (2023), EPC lecture notes, Univ. of Pisa · ai4aging.org